锐驰智光
RICHBEAM

RBView Use Manual (V1.0.1.1)

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Revision instructions

Based on the experience of all users, this manual is constantly updated without further notice. Please be sure to obtain the latest version of this manual through the official public network (http://www.richbeam.com) before using the product. Please contact us if having any questions about this manual.

user feedback:<support@richbeam.com>

Revision History

Date	Version	Release Content
2022-06-15	1.0.1.0	Initial Release
2022-08-25	.0.11.1	Change software usage library documents, and add CSV saving function

Summary

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This document describes the installation, functions and usage of RBView software, and it will be updated with the functional iteration of LakiBeam series products. RBView is only used to control and view data of LakiBeam series products, and cannot be transferred to other uses.

1 System Configuration

Operating system: Windows10 (64-bit) and above/Linux

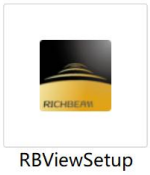
Memory: Greater than 4GB

CPU: Quad-Core 2.0 GHz

2 installation

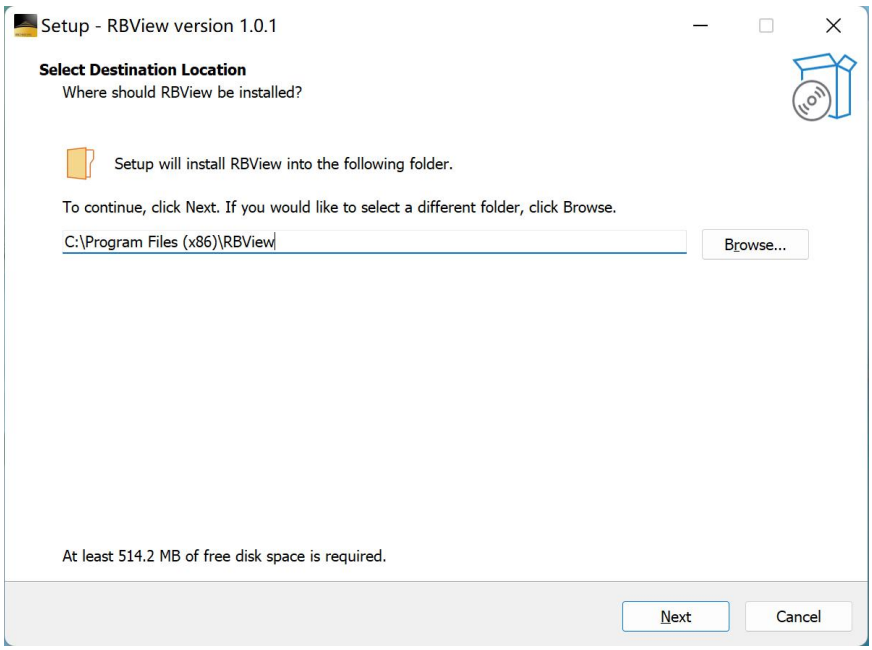
2.1 Program installation

Setup program RBViewSetup.exe

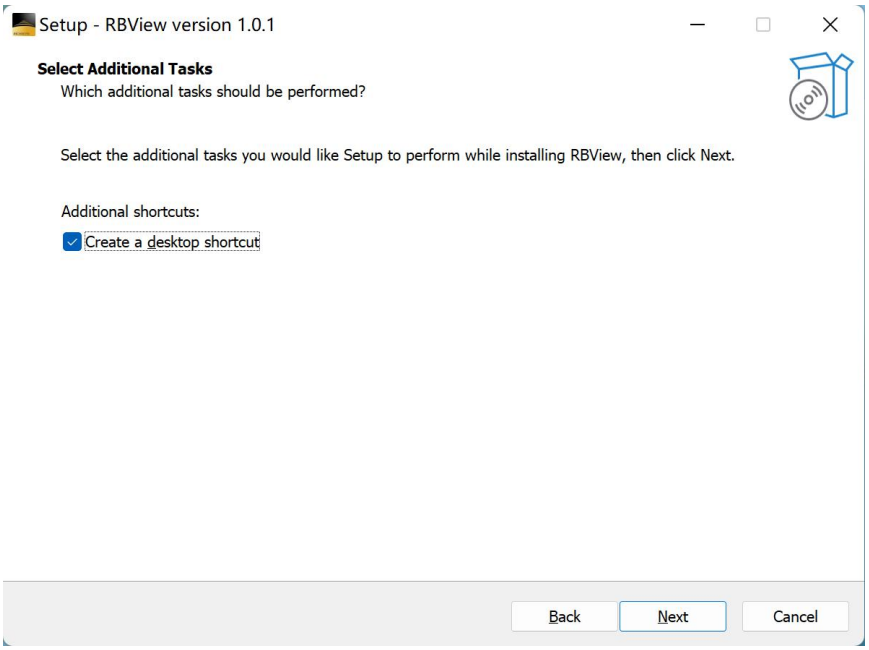


Installation Process:

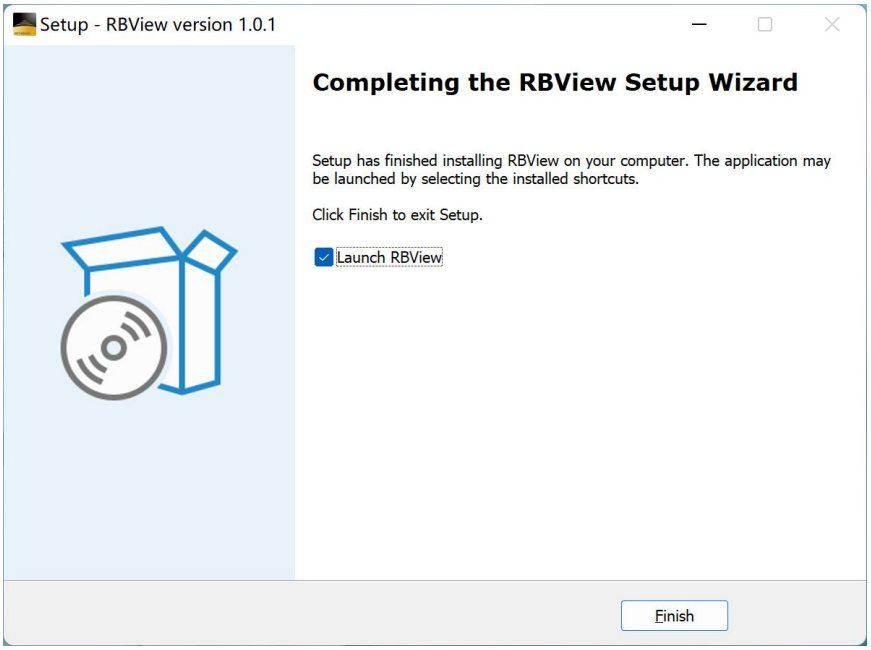
1. Double-click on "RBViewSetup.exe" to start the installation. After selecting the installation destination, click "Next" to proceed.



2. After creating the desktop shortcut, click "Next" for the next step



3. Click on the "Finish" to complete the installation process, and you're ready to use the RBView



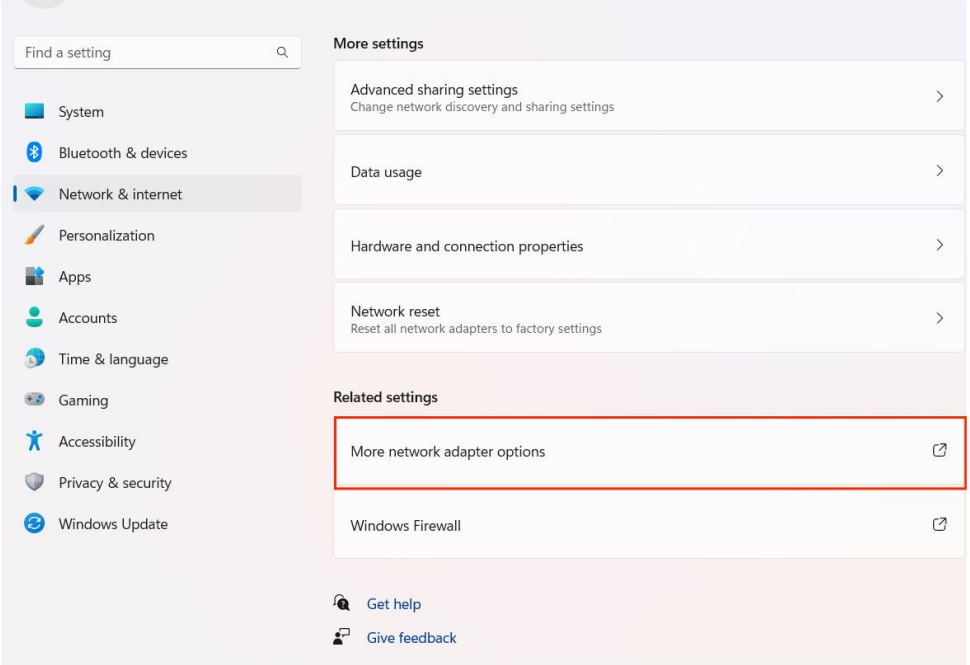
2.2 Network setup (network cable connection)

LakiBeam1(L) starts to work after being powered on, and starts to collect and transmit data .The default network setting of the radar is static mode, in which the radar IP address is 192.168.198.2, and the subnet mask is 255.255.255.0.

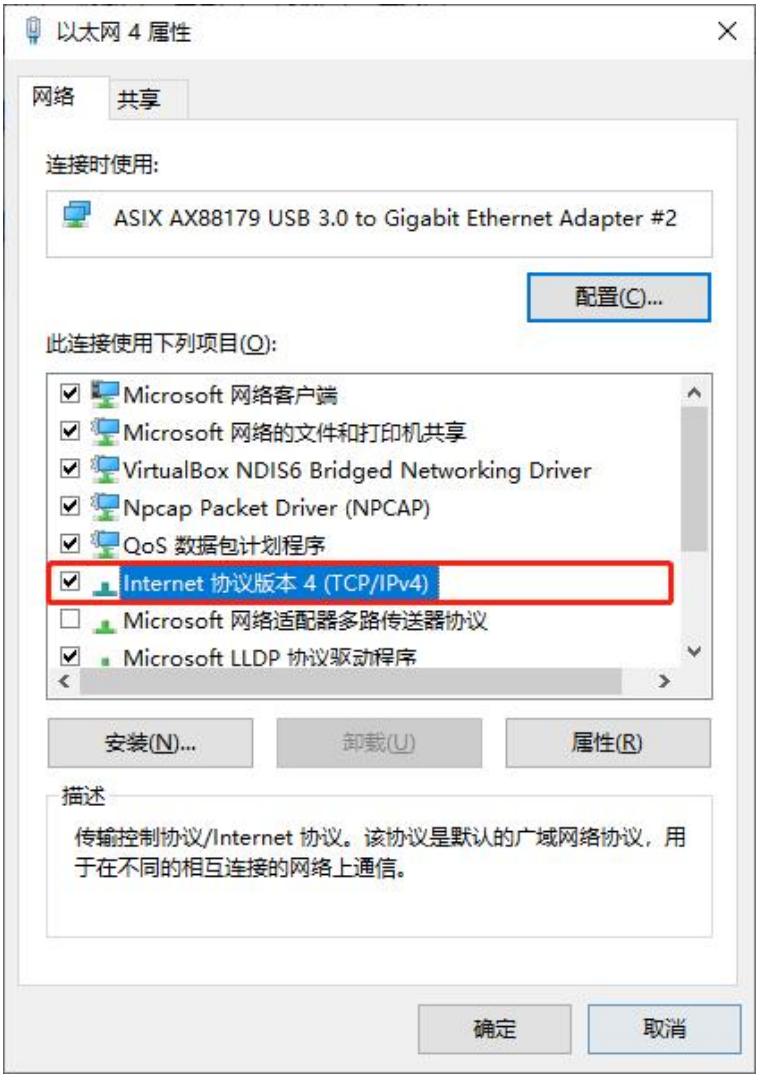
The IP address of the computer (or other device connected to the radar) is 192.168.198.1, and the port number is 2368.

When you use the RBView to view the point cloud data output by the radar,you need to configure the network card that connects the radar to the computer device.

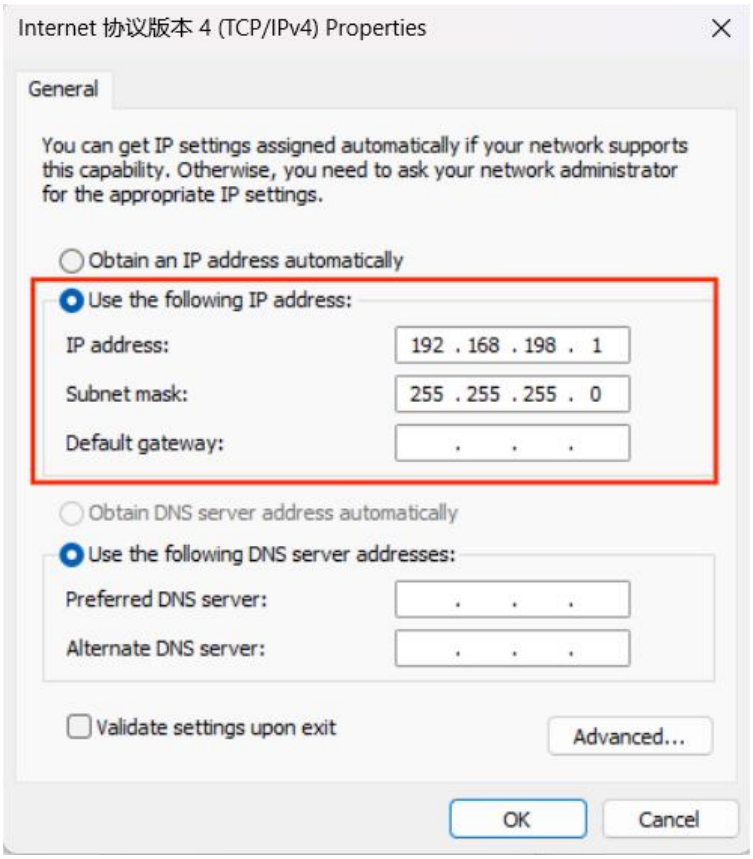
1. Open "Network & Internet", and select the “More network adapter options”



2. Find the network card connecting the radar and the computer device, right-click, select the properties, and double-click the "Internet Protocol Version 4 (TCP / IPv4)"



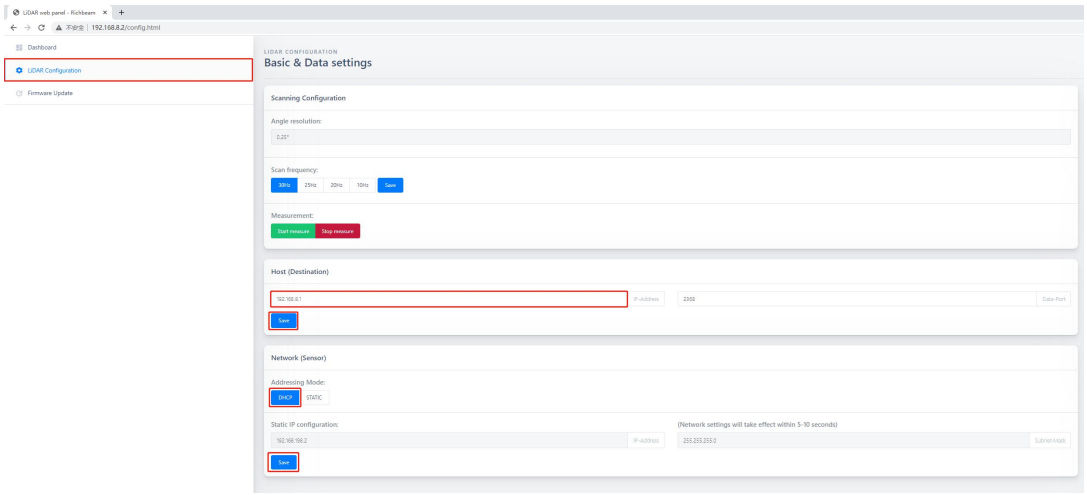
3. Click the tab "Use the following IP address" tab to modify the IP address to 192.168.198.1 and the subnet mask to 255.255.255.0, and click"OK"



2.3 Network Settings (USB connection)

When the device is connected to the computer through the USB Type-C data cable, it takes about 30 seconds for the device to complete the startup and the self-test after the connection is properly established, and the computer will recognize a USB mass storage device and an RNDIS network device (Windows/Linux) after the self-test is completed. At this time, there is no need to configure the RNDIS device, because an IP address of 192.168.8.1 is fixedly allocated to the virtual network card through the built-in DHCP server in the device, and you need to set up the network by visiting http://192.168.8.2 using your computer's web browser to access the built-in web server. The specific settings are as follows:

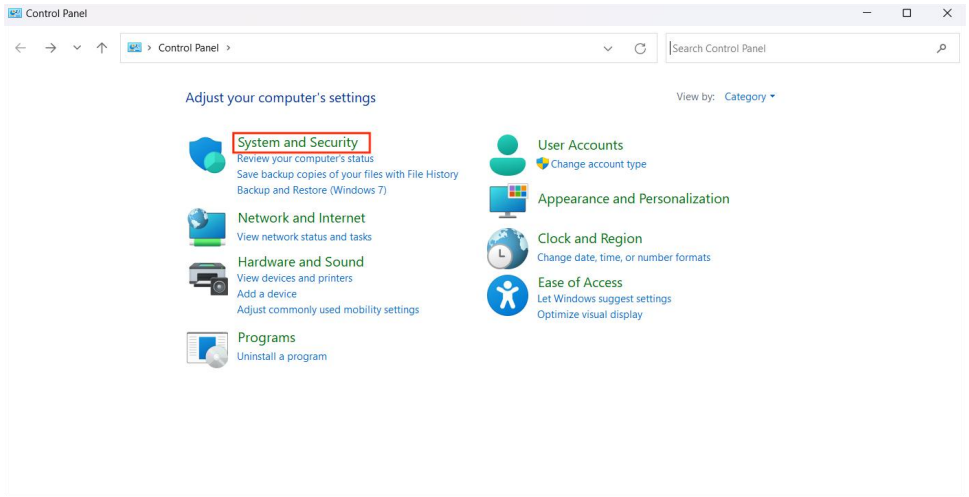
Click the "LiDAR Configuration" tab to enter the Radar Configuration Page, modify the IP address of Host (Destination) to 192.168.8.1, and click "Save" to save the File, change the Network Mode to DHCP, and click "Save" to save the File, then the new configuration will take effect after 5-10s.

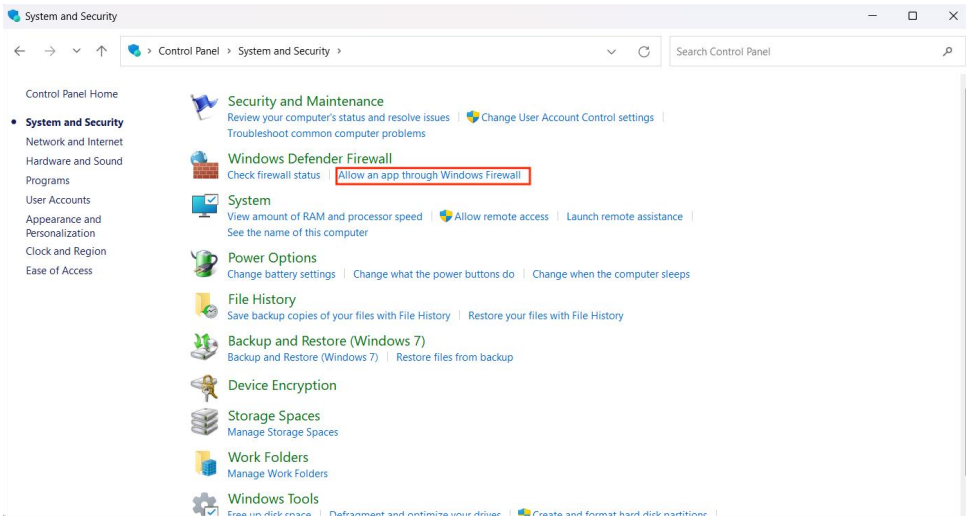


2.4 Firewall Settings

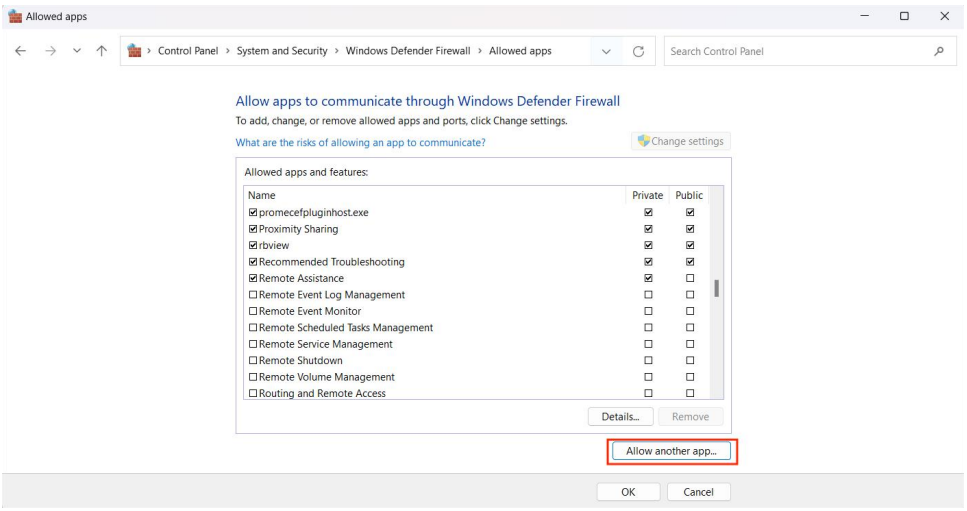
Since LakiBeam1(L) connects with the RBView through a network interface and continuously sends UDP data packets to a computer, the LakiBeam1(L) needs to penetrate a firewall of an operating system to obtain the data,so the firewall should be set up before use.

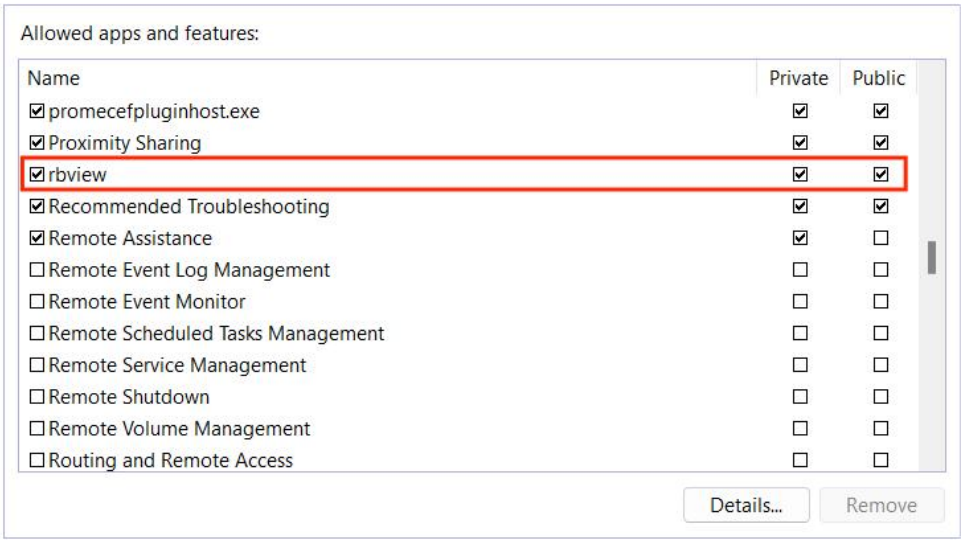
1. Open the Control Panel-> System and Security-> Allows an app through Windows Firewall





2. Click "Change Settings" to add the path of RBView.exe,which allows the RBView to communicate via the Windows firewall,click"OK"to complete the setup.

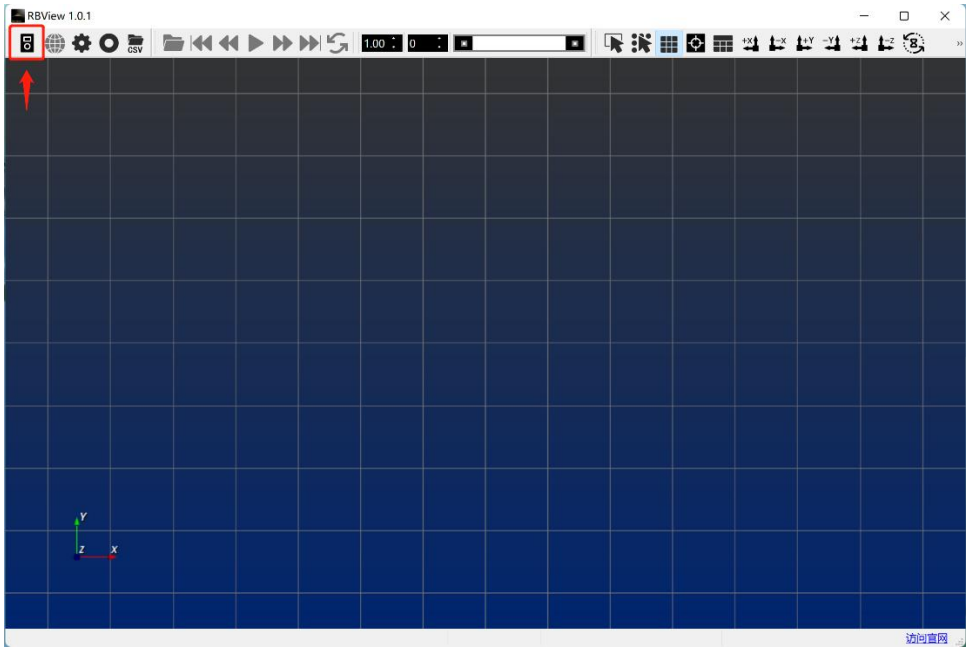




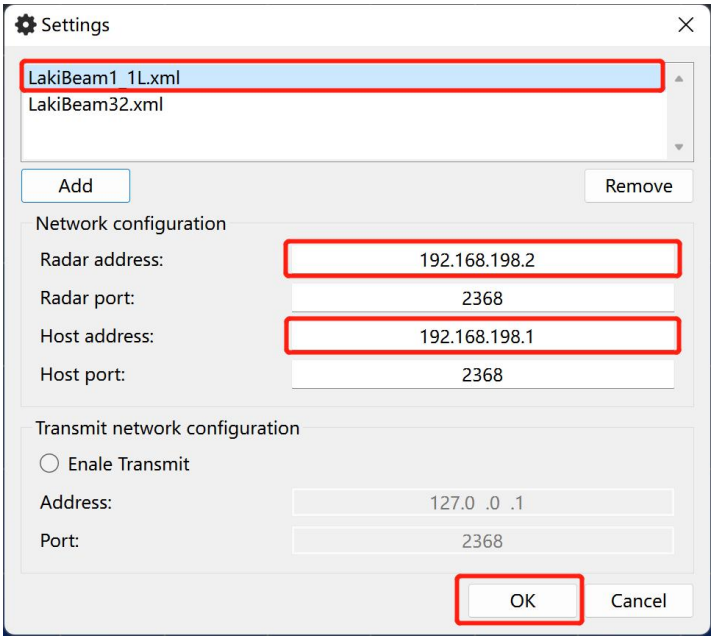
3 Use of RBView

3.1 Acquisition radar datar

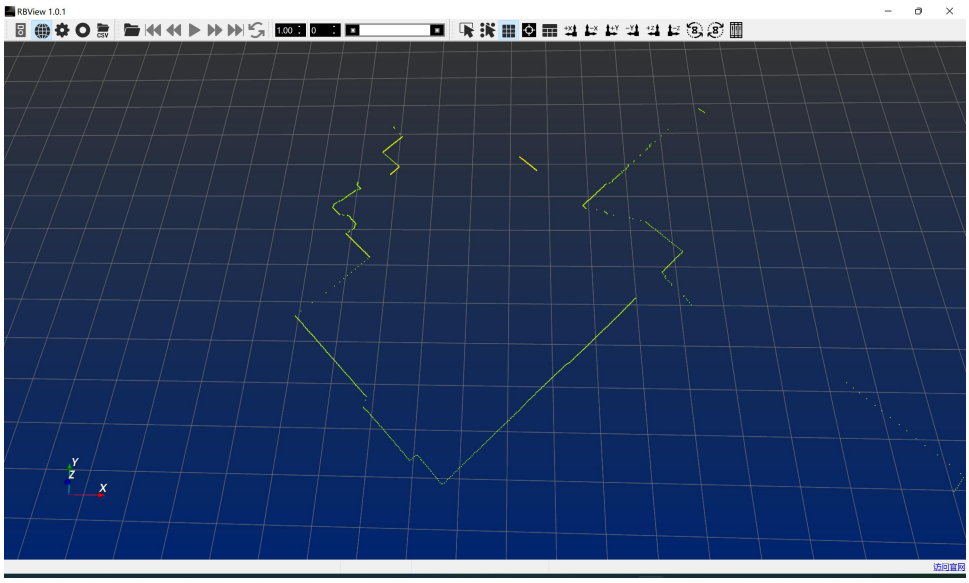
Click the"Open radar configurations" icon:



Select the corresponding xml file of LakiBeam1 (L) in the Settings tab that pops up,and confirm that the settings of the radar IP and Host IP in the"Network configuration"are consistent with the radar Web terminal setting, click"OK":



Click the"network"icon to view the point cloud data, as shown in the figure, and click the network icon again to pause:



The basic operations of viewing a point cloud are:

Move with Left Mouse Button: Rotate the Coordinate System

Drag/Rotate Wheel with Right Mouse Button: Zoom in / out Coordinate System and Overall Point Cloud

Drag with Mouse Wheel: Pan Coordinate System and Overall Point Cloud

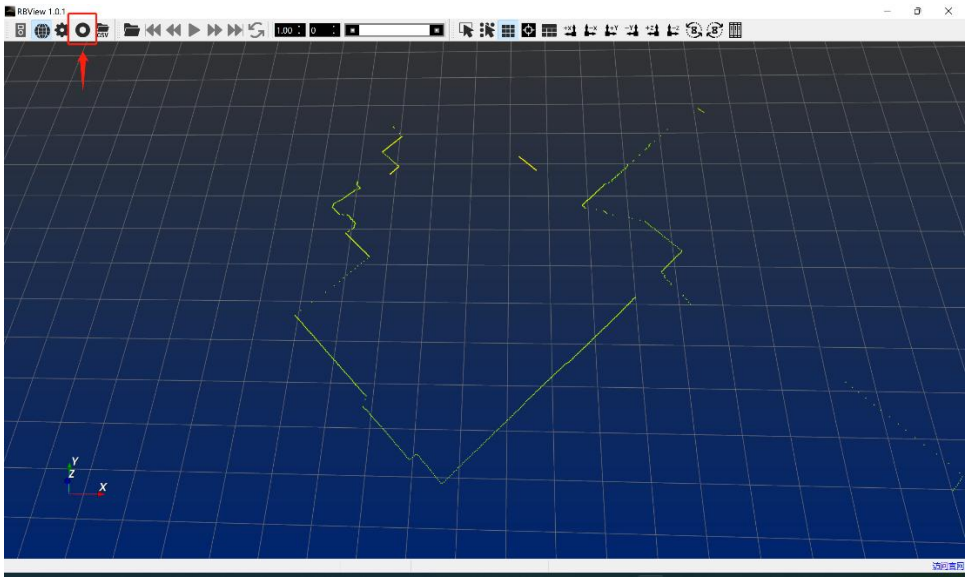
Plus/Minus: Zoom in/out Point Cloud Thickness

Tap Axis in Bottom Left: Zoom in/out, Drag to Move

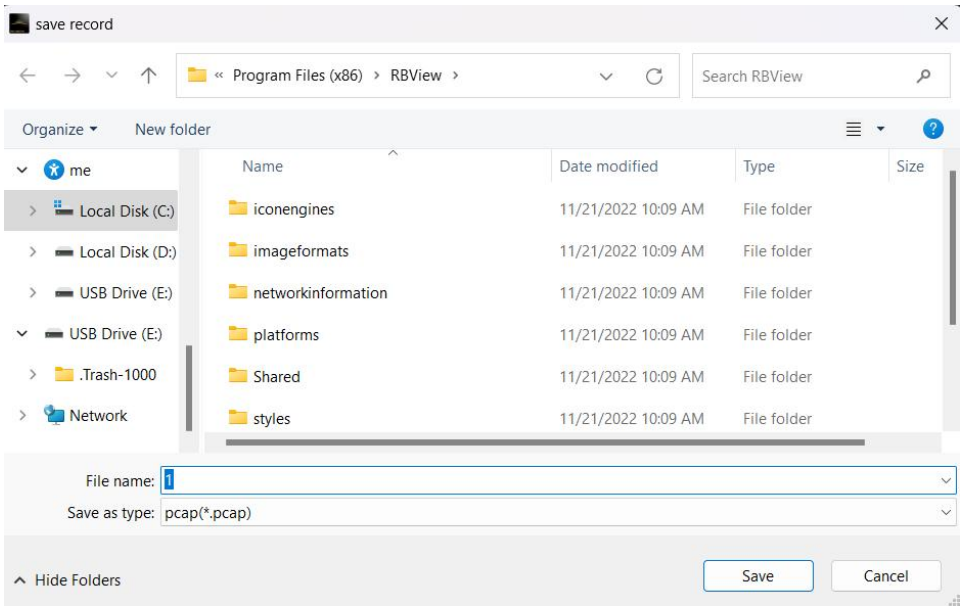
3.2 Point cloud data recording

When collecting point cloud data, RBView can record and save the collected data in real time, so as to facilitate users to further view and analyze the collected data.The recording process is as follows:

1. Click on the Recording button:



2. Select the save location and name the file:



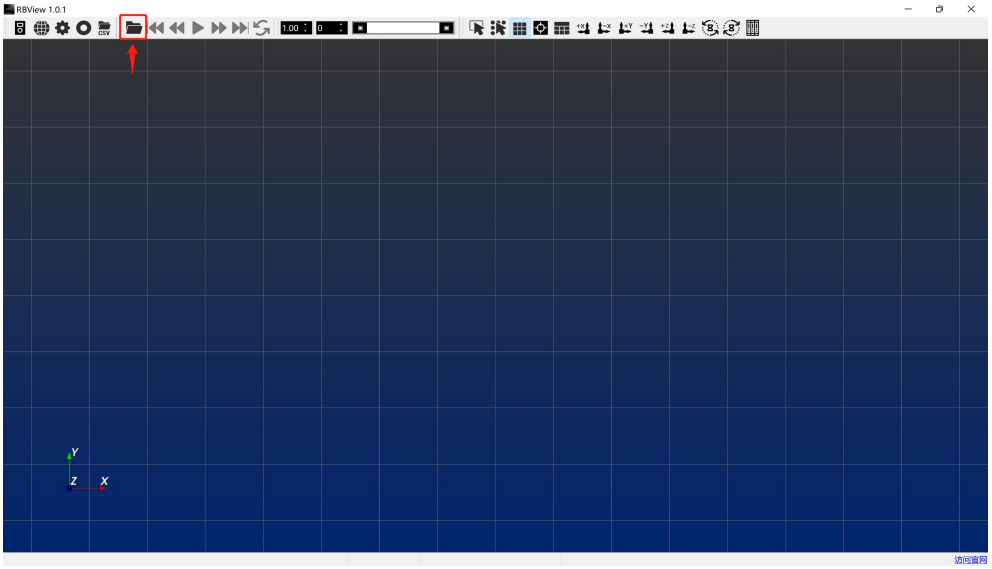
3. Click on "Save" to start the recording

4. Click the recording button again to stop the recording

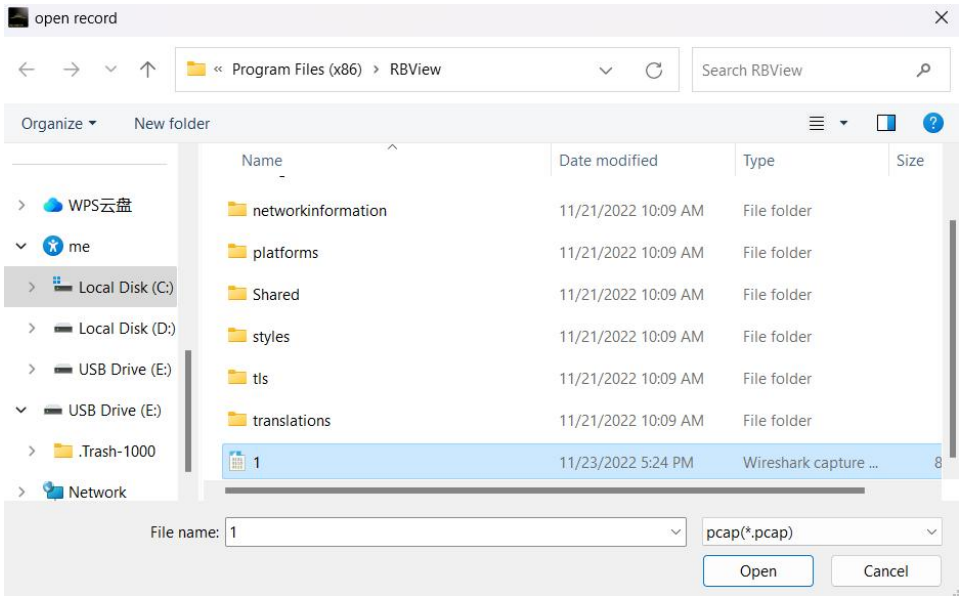
3.3 Play the point-cloud data

After recording the point cloud data, RBView can load the recorded data for playback, facilitate users to better analyze the data. The playback process is as follows:

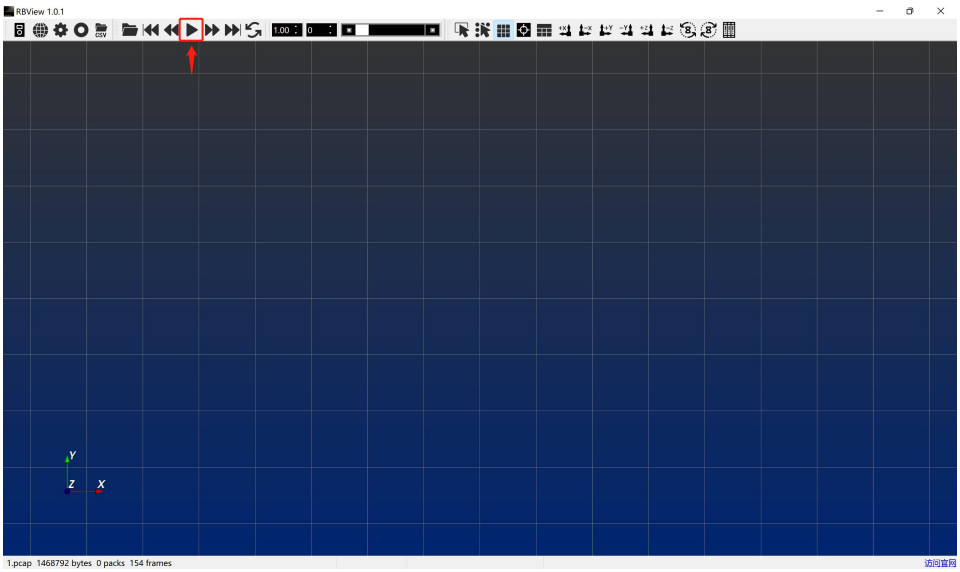
1. Click "the Open radar configurations " icon, and configure the Settings tab, click the Play button:



2. Select the recorded data and click "Open" to load the point cloud data file:



3. Click the Play button to start playing:



The buttons shown below from left to right, are: first frame, previous frame, play/pause, next frame, last frame, loop.

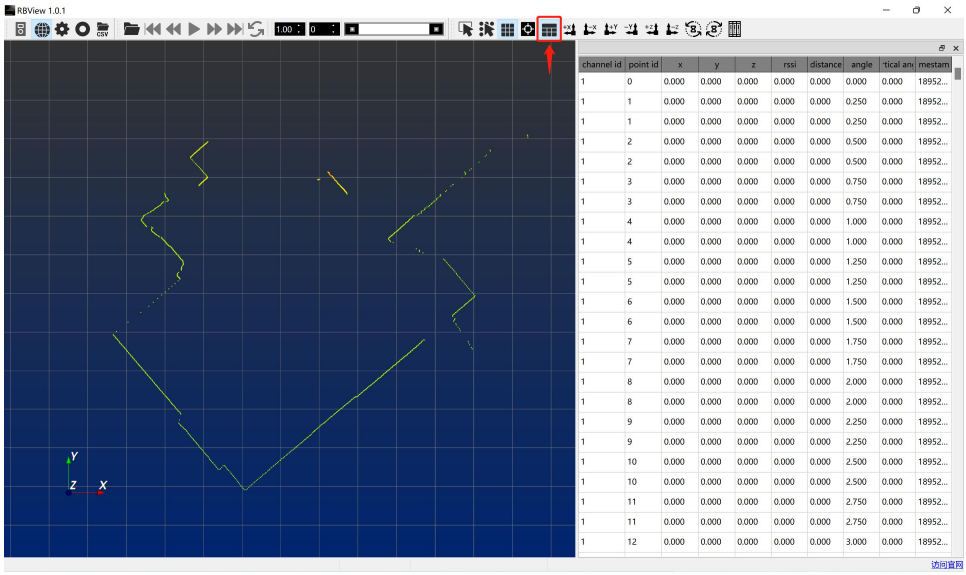


The following figure represents the progress bar when the point-cloud data plays:



3.4 Detailed data viewing

In order to allow users to see the actual radar data intuitively, RBView provides a side table display function. Users can view the display information of any depth data by opening the table. By clicking the table button, the table displaying all the point cloud data information will pop up, as shown in the figure below:

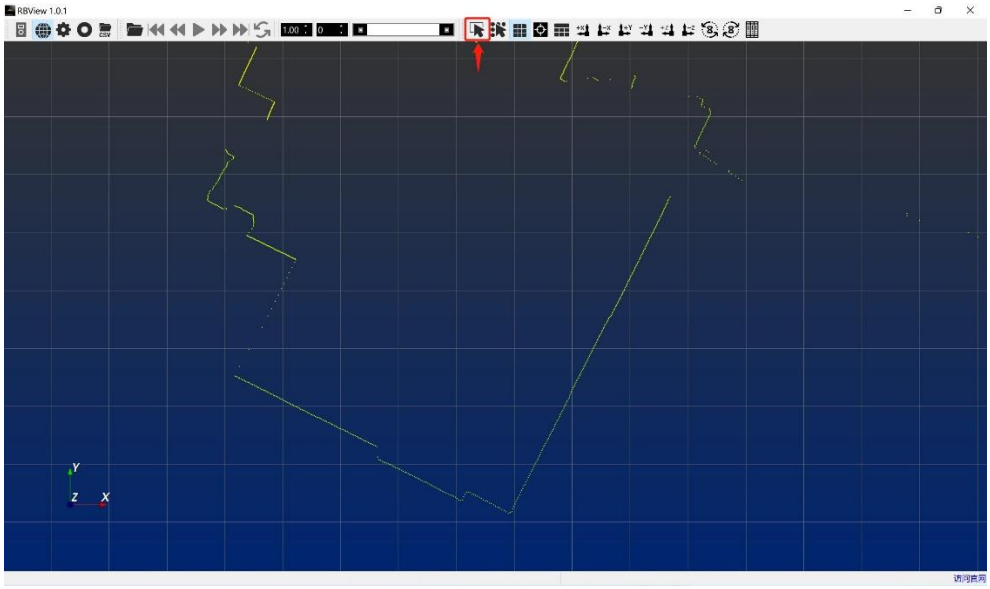


The information showing the point cloud in the table includes: channel ID (both Lakibeam1 and Lakibeam1L channels are 1), point ID, XYZ coordinates under the Cartesian coordinate system, horizontal angle, distance from the measured object to the origin of the radar, signal intensity, and timestamp.

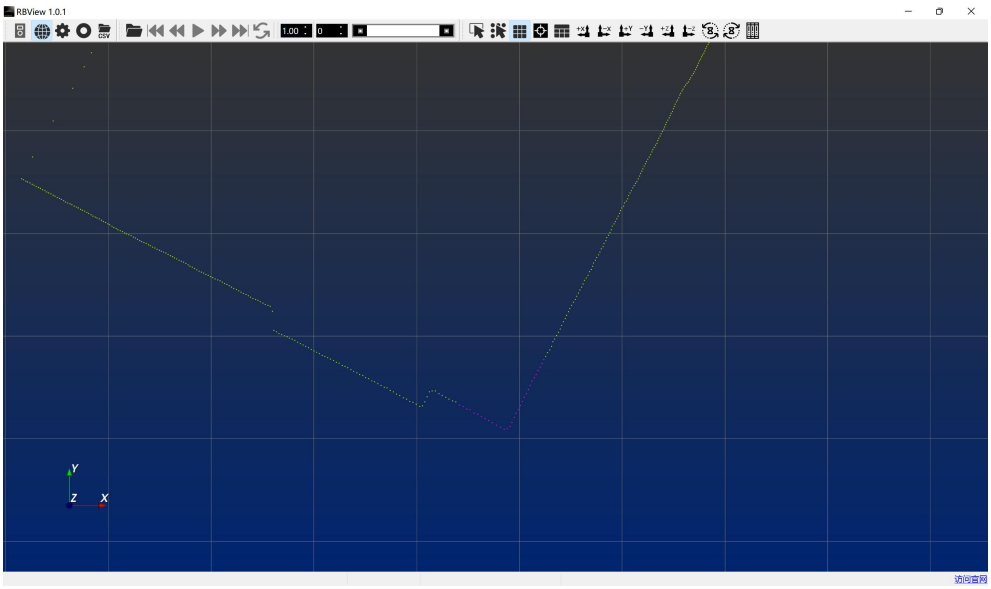
3.5 Region selection

RBView provides a region selection function that allows users to view the point cloud of a certain region and exclude the interference of other points. Specific operations are as follows:

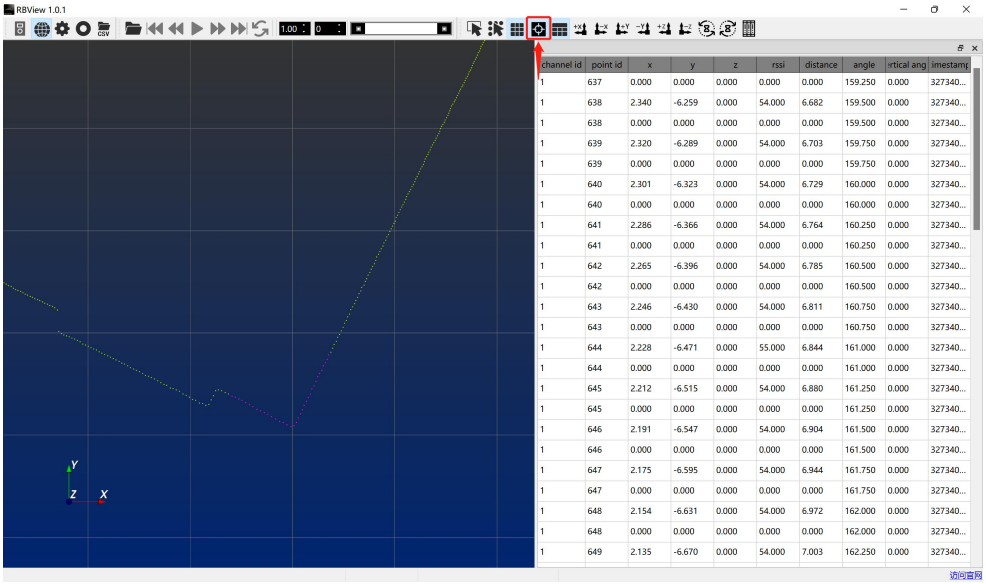
1. Click on the region selection button:



2. Box Select some of the point clouds that you want to view:



3. Open the side display table, and click the "tables show select" button:



Region selected data is highlighted (in red) and can be observed in the side table. Uncheck the region can use the keyboard "Esc", if the user places the mouse over the Point Cloud Display Interface, or use the Toolbar area to unselect the"disselect" button, as shown in the figure below:

